

Part 2 [Set Theory & Algebra]

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- Power set $P(A)$ is set of all possible subset of A .
- $|P(A)| = 2^{|A|}$
- Symmetric difference (XOR): $A \oplus B = (A - B) \cup (B - A)$
- Idempotent law: $A \cap A = A \cup A = A$
- Absorption law: $A \cup (A \cap B) = A$
- $A \cap P(A) = \emptyset$, $P(A) \cap P(P(A)) = \{\emptyset\}$
- Sets are commutative and associative over $(\cup, \cap, \oplus)$ but distributive over $(\cup, \cap)$ only.
- No. of finite relation possible over cartesian set = $2^{(m \times n)}$
- Reflexive: Comparison among same item of set (x, x) .
Smallest reflexive $|R_n| = n$, largest reflexive $|R_n| = n^2$
No. of reflexive relation possible = $2^{n^2 - n}$
No. of irreflexive relⁿ possible = $2^{n^2} - 2^{n^2 - n}$
- If R_1 and R_2 are reflexive then $(R_1 \cup R_2)$ & $(R_1 \cap R_2)$ both are reflexive but if R_1 and R_2 are irreflexive then $(R_1 \cup R_2)$, $(R_1 \cap R_2)$ & $(R_1 - R_2)$ all are irreflexive.
- Reflexive: all the diagonal elements are necessary.
Irreflexive: even a single diagonal element doesn't require.
- There is no relation which is both reflexive & irreflexive.
No. of irreflexive relation = $2^{n^2} - 2^{n^2 - n}$, Smallest = $\{\emptyset\}$,
largest = $n^2 - n$.
- Symmetric: If (aRb) then (bRa)
Smallest relation = $\{\emptyset\}$, Largest relⁿ = n^2
No. of symmetric relations = $2^n \cdot 2^{\frac{n^2 - n}{2}} = 2^{\frac{n(n+1)}{2}}$
 $|Sym \cap Reflexive| = 2^{\frac{n^2 - n}{2}} = |Sym. \cap Irr|$
- Antisymmetric (As): If (xRy) and (yRx) then $x = y$.
 $|As| = n + \frac{n^2 - n}{2}$
Smallest = n , Largest = $|A \times A| = n^2$
- Symmetric relations are closed under $(\cup, \cap, -)$
 $n(S \cap I) = 2^{\frac{n^2 - n}{2}}$
 $n(As) = 2^n \cdot 3^{\frac{n^2 - n}{2}}$
 $n(As \cap S) = 2^n$
 $n(R \cap As) = 3^{\frac{n^2 - n}{2}}$
 $n(R \cap As) = 3^{\frac{n^2 - n}{2}}$

Asymmetric: Stricter version of Anti-symmetric.

$$|Asy| = 3^{\frac{n^2-n}{2}}, \text{ Smallest} = 4, \text{ Largest} = \frac{n^2-n}{2}$$

- Reflexive relⁿ can't be antisymmetric Asymmetric bcoz of no diagonal elements
- Every asymmetric relation is irreflexive and anti-symmetric.
- It is closed under subset, intersection and set difference operation.

Transitivity: If (xRy) & (yRz) then (xRz)

→ It is closed under '∧' but not '∨'.

Equivalence Relⁿ: Closed under '∧' but not '∨'.

→ Reflexive, Symmetric & Transitive together

Partial Order Relation: Reflexive, transitive, Anti-symmetric together

Totally Ordered Set: Linear order set or chain. It means relation exist b/w every pair of elements

- Representation of poset: Hasse diagrams
- Don't need to show reflexive (loop) or transitive relation in Hasse diagram.

Maximal element → No relation above elements

Minimal element → No relation with below elements

Maximum → Maximum in all the maximal.

Minimum → Minimum in all the minimal.

Upper bound / Lower bound: In a given hasse diagram either go only up or only down but not both up-down.

GLB or LUB: GLB: (Meet, ∧), LUB: (Join, ∨, Supremum)

Join Semi Lattice: It consist LUB for each pair of vertices.

Meet Semi Lattice: ——— GLB ———

Lattice → If hasse diagram is both Join semi Lattice and meet semi Lattice

Properties → follows commutative, associative, but not distributive

$$a \cup (b \cap c) = (a \cup b) \cap (a \cup c) \quad \leftarrow \text{of lattice follow these called}$$
$$a \cap (b \cup c) = (a \cap b) \cup (a \cap c) \quad \leftarrow \text{distributed lattices}$$

Sub lattice: It has same LUB and GLB as parent lattice. (34)

→ If '1' is Upper bound & '0' is lower bound, then

$$LUB(a \text{ and } 1) = a \vee 1 = 1, LUB(0, a) = a \vee 0 = a$$

$$GLB(a, 1) = a \wedge 1 = a, GLB(a, 0) = 0$$

→ Bounded lattice is a finite lattice because it has both the upper bound and a lower bound.

→ for Infinite lattice we can't find LB, UB.

→ for any set UB is Universal Set and LB is ϕ .

Complement of element in lattice: $a \vee a^c = 1, a \wedge a^c = 0$

Pair of element with same GLB and LUB called as complement.

- in distributive lattice every element have atmost 1 complement
- in complemented lattice every element have atleast 1 complement
- in boolean algebra every element have only 1 complement

Groups:

Algebraic structure: Each set is closed under some operations.

Semi Group: If algebraic structure is associative. $\{(\mathbb{Q}^+, *)\}$

Monoid: Semi-group with identity element.

for union operation ϕ works as identity element.

Groups: Monoid with inverse element.

$[(\mathbb{Q}^+, *)]$ is a group but $(\mathbb{Q}, +)$ doesn't. Similarly $(\mathbb{P}(\mathbb{A}), \cup)$ is also not a group.

Abelian Group: It is also called commutative group. Inverse of 'e' is 'e' itself. Unique inverse & identity element.

for strings identity element $e = \epsilon$.

→ Identity element: $a * e = a$, Inverse element: $a * a^{-1} = e$
 $*$ → It can be any operation.

Finite Group: Group with finite no. of elements

→ In modulo 'm' operations range of number should be $(0 \rightarrow m-1)$.

→ Order of an element 'a' is 'n' if $a^n = e$ for smallest value of 'n'.

→ Order of identity element is always 1.

→ Order of a, a^{-1} are same and order of element always divides order of group.

Subgroup: e and G are trivial subgroups. All other are proper subgroups.

→ If 'H' is a subgroup of 'G' then $O(G)/O(H) = k$
If it divides then only we will check for subgroup further.
After applying divisor rule verify subgroup should be closed under group's operation.

→ Subgroups are closed under 'n' but not under 'v'.

→ Every subgroup of abelian group is also abelian group.

Cyclic group: If every element of group can be written in form of (a^n) then $a \rightarrow$ generating element.

Number of generator = $\phi(n)$ means no. of elements $< n$ which are relatively prime to n .

→ Properties: If $n = pq$ then $\phi(n) = \phi(p) \cdot \phi(q)$
where, p, q are distinct & prime.

② $\phi(p) = p-1$ for $p \rightarrow$ prime.

③ $\phi(p^n) = p^n - p^{n-1}$

→ If $(G, *)$ is a cyclic group with generator a then,

① a^{-1} is also a generator

② $O(a) = O(G)$

→ Every cyclic group is abelian, every group of prime order is cyclic.

→ If $|X| = m, |Y| = n$ then no. of fn possible = n^m .

→ One-One fn: $|Y| \geq |X|$: no. of fn = $n P_m$

If fn is bijective $|X| = |Y|$ and it is one-one and onto then,
no. of one-one = no. of onto = $n!$

if $|X| \neq |Y|$ then only onto possible. So

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no. of onto $f = n^m - nC_1(n-1)^m + nC_2(n-2)^m - \dots + (-1)^{n-1} nC_{n-1}(1)^m$

→ Two irrational no. x & y can give $(x+y)$ as rational.

→ $f(A \cup B) = f(A) \cup f(B)$, $f(A \cap B) \neq f(A) \cap f(B)$

$f(A \cap B) \subseteq f(A) \cap f(B)$. Equals when both are one-one.

→ ~~Relation with empty set is symmetric and transitive but not reflexive.~~

→ $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$, $\phi \in P(A)$, $\phi \subseteq P(A)$

→ if $h = f \circ g$ then if $h \Rightarrow$ onto then $f \Rightarrow$ onto.

→ N^N is countable but K^N is not countable.

→ Complemented lattice is proper subset of bounded lattices.

→ Set of all rational no. is countable.

→ A finite lattice is always bounded because it has both LUB and GLB (due to lattice).

→ if a relation is defined on power set of some set then,
 $a \cap a = \phi$ for $[a = \phi]$

→ Empty relation over finite set is not reflexive but symmetric & transitive.

→ Empty relation over empty set is an equivalence relⁿ.

→ Every finite lattice has a least element and max. element.

→ Every poset doesn't have a least or greatest element (ex: Bipartite graph).

→ There is just a single relation which is both equivalence and partial order relation. → (Identity relⁿ)

→ No. of integers between $(1 \rightarrow n)$ divisible by 'K' is given by

$$\left\lfloor \frac{n}{k} \right\rfloor \rightarrow \text{floor } \frac{n}{k}$$

Combinatorics

Generalized Pigeonhole: If n -objects are placed into k -boxes then atleast one box contains $\lceil n/k \rceil$ objects.

~~$\rightarrow nC_r = nC_{n-r}$~~ , $nC_r + nC_{r-1} = n+1C_r$

$\rightarrow (x+y)^n = nC_0 x^n + nC_1 x^{n-1}y + \dots + nC_n y^n$

General term: $nC_r x^{n-r} y^r$

$\rightarrow \sum_{k=0}^n nC_k = 2^n$, $\sum_{k=0}^n (-1)^k nC_k = 0$

Soln of linear homogeneous recurrence relation.

① If eqn has same roots as ' α ' then,

$$a_n = \alpha_1 a^n + \alpha_2 n a^n + \alpha_3 n^2 a^n + \dots$$

② If eqn has different roots then,

$$a_n = \alpha_1 a^n + \alpha_2 b^n + \alpha_3 c^n + \dots$$

If $G(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_k x^k = \sum_{k=0}^{\infty} a_k x^k$

Useful generating fn

$$(1+x)^n = nC_0 x^0 + nC_1 x^1 + \dots + nC_n x^n = \sum_{k=0}^n \frac{nC_k x^k}{a_k}$$

$$(1+ax)^n = nC_0 a^0 x^0 + nC_1 a x + \dots + nC_n a^n x^n = \sum_{k=0}^n \frac{nC_k a^k x^k}{a_k}$$

$$\frac{1-x^{n+1}}{1-x} = 1+x+x^2+\dots+x^n = \sum_{k=0}^n x^k$$

$$\frac{1}{1-x} = 1+x+x^2+\dots = \sum_{k=0}^{\infty} x^k$$

~~$\frac{1}{1-ax} = 1+ax+a^2x^2+\dots = \sum_{k=0}^{\infty} a^k x^k$~~

$$\frac{1}{1-x^r} = 1+x^r+x^{2r}+\dots = \sum_{k=0}^{\infty} x^{rk}$$

$$\frac{1}{(1-x)^2} = 1+2x+3x^2+\dots = \sum_{k=0}^{\infty} (k+1)x^k$$

~~$\frac{1}{(1+x)^n} = 1 - nC_1 x + n+1C_2 x^2 + \dots = \sum_{k=0}^{\infty} n+k-1C_k (-1)^k x^k$~~

$$\frac{1}{(1-x)^n} = 1 + nC_1 x + nC_2 x^2 + \dots = \sum_{k=0}^{\infty} n+k-1 C_k x^k \quad (36)$$

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots = \sum_{k=1}^{\infty} \frac{(-1)^{k+1} x^k}{k}$$

① $G(x)$ for $\{1\} = \frac{1}{1-x}$ ② $G(x)$ for $n = \frac{x}{(1-x)^2}$

$$\rightarrow nC_r = \frac{n!}{r!(n-r)!}, \quad -nC_r = \frac{(-1)^r (n+r-1)!}{r!(n-1)!} = \boxed{n+r-1 C_r}$$

$\rightarrow$ No. of derangement $D_n = \sum_{i=2}^n (-1)^i \frac{n!}{i!}$ $\rightarrow$ Nothing can be placed in right box.

$\rightarrow$ Generating fn for fibonacci series = $\frac{1}{1-x-x^2}$

$\rightarrow$ No. of substrings formed from a length of string 'n' is -

$$\boxed{\frac{n(n+1)}{2} + 1}$$

$\rightarrow$ No. of ways to divide 'k' things among 'n' persons $\rightarrow$ Similar $\rightarrow$ different things

$$one = \boxed{n+k-1 C_k}$$

$\rightarrow$ No. of moves required on travelling (i, i) to (d, m)

$$= \frac{(\text{Total no. of blocks to travel})!}{(\text{Total UP})! (\text{Total Right})!}$$

$\rightarrow$ If there are 'n' objects of 'k' types such that all items occur with some frequency then,

No. of ways of arranging them = $\frac{n!}{(f_1! f_2! \dots f_k!)} \times (k!)$

Here, $f_1 + f_2 + \dots + f_k = n$

if values f_i are all different

$\rightarrow nC_1 + 2 \cdot nC_2 + 3 \cdot nC_3 + \dots + n \cdot nC_n = \sum_{k=1}^n k \cdot nC_k = \underline{n \cdot 2^{n-1}}$

Counting Principles

- ① Product Rule: If there are n tasks that can be done in $n_1, n_2, \dots, n_n$ ways then total no. of ways to do the procedure is $n_1 \times n_2 \times \dots \times n_n$.

- ② Sum Rule: If task can be done either in n_1 ways or n_2 ways then total no of ways = $n_1 + n_2$.

- ③ No. of f^n : If $|\text{domain}| = |m|$ and $|\text{codomain}| = |n|$ then no. of f^n possible = n^m

for one-one functions:- $n \geq m$ and No. of fn = 'k',

$$k = \frac{n(n-1)(n-2) \dots (n-k+1)}{k!}$$

Ex → A password is 6-8 characters long where each character is an uppercase letter or a digit. Each password must contain at least one digit. No. of password is —.

Sol: No. of password = $16 + 17 + 18$

$$P_6 = 36^6 - 26^6, P_7 = 36^7 - 26^7, P_8 = 36^8 - 26^8.$$

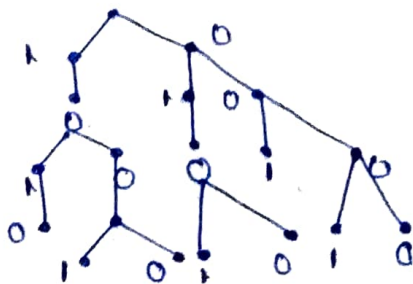
Subtraction Rule: If a task can be done in either n_1 or n_2 ways then total no. of ways possible is $n_1 + n_2 - (n_1 \cdot n_2)$.

Ex → How many bit strings of length 8 either start with bit 1 or end with 00.

Ans $\rightarrow 2^7 + 2^6 - 2^5$

Division Rule : (Not important)

Tree Diagrams : Bit string of length 4 don't have consecutive 1's



Pigeonhole: To keep $k+1$ objects in k boxes, there is at least 1 box with two or more objects.

Collary: A function 'f' from a set with $(k+1)$ or more elements to a set with 'k' elements is not one to one.

Generalized Pigeonhole: If N objects are placed into k objects boxes, then at least one box contains $\lceil N/k \rceil$ objects.

Permutations $\rightarrow$ No. of ordered arrangements

Ex $\rightarrow$ Let $S = \{1, 2, 3\}$ then $3, 1, 2$ is permutation of S and $3, 2, 1$ is a 2-permutation of S .

$\rightarrow$ If n, r are positive integers such that $r \in [1, n]$ then, there are $P(n, r) = n(n-1)(n-2)\dots(n-r+1)$, r -permutations of a set with n -distinct elements.

$\rightarrow$ No. of permutations of letters ABCDEFGH contains the string ABC equals to $6!$. (Hint: Pair ABC as a single element.)

Combinations: No. of r -combinations of a set with n -distinct elements is denoted by $C(n, r)$ or $\binom{n}{r}$ and called as binomial-coefficients.

$$nC_r = nC_{n-r}, \quad nC_r + nC_{r-1} = n+1C_r$$

Binomial Theorem:

$$(x+y)^n = nC_0 x^n + nC_1 x^{n-1}y + \dots + nC_n y^n$$

$\rightarrow$ Coefficient of $x^{12}y^{13}$ in,

$$(1) (x+y)^{25} = \frac{25C_{13}}$$

$$(2) (2x-3y)^{25} = \frac{25C_{13} \times 2^{12} (-3)^{13}}$$

$$nC_r x^{n-r} y^r$$

$\rightarrow$ If n is a non-negative integer then,

$$\sum_{k=0}^n \binom{n}{k} = 2^n$$

$$\text{or } \sum_{k=0}^n (-1)^k \binom{n}{k} = 0$$

$$\text{or } \sum_{k=0}^n 2^k \binom{n}{k} = 3^n$$

Vandermonde's Identity

$$\frac{n+m}{r} C_r = \sum_{k=0}^r \binom{m}{r-k} \binom{n}{k} \quad \text{--- (1)}$$

20 (00) $2n C_n = \sum_{k=0}^n \binom{n}{k}^2$, at $m=r=n$ in eqn (1)

(00) $n+1 C_{r+1} = \sum_{k=r}^n \binom{k}{r}$

Permutation with Repetition: The no. of r -permutations of a set of n -objects with repetition allowed is n^r .

Combination with repetition: There are $C(n+r-1, r)$ r -combinations from a set with n -elements when repetition of elements is allowed.

Permutation with indistinguishable Object:

Ex: How many strings can be made by reordering the letters of word SUCCESS?

Solⁿ: $\frac{7!}{1! 2! 1! 1! 1!}$

✓ Distributing objects into boxes:

Ex: How many ways are there to distribute hands of 5 cards to each of four players from the standard deck of 52 cards.

Solⁿ: $\frac{52!}{47! 5!} \cdot \frac{47!}{42! 5!} \cdot \frac{42!}{37! 5!} \cdot \frac{37!}{5! 32!} = \frac{52!}{5! 5! 5! 5! 32!}$

Recurrence Relation and Generating function

The recurrence relation $P_n = (1, 1) P_{n-1}$ is a linear homogeneous recurrence relation of degree 1 and $f_n = f_{n-1} + f_{n-2}$ is linear and having degree 2.

→ $A_n = a_{n-1} + a_{n-2}$: not a linear.

$H_n = 2H_{n-1} + 1$: not Homogeneous

$B_n = nB_{n-1}$: no constant coefficients

→ Solving Linear Homogeneous Recurrence Relation with constant coefficients

Ex-1 $a_n = a_{n-1} + 2a_{n-2}$ with $a_0 = 2$ and $a_1 = 7$. find the Solⁿ.

Solⁿ: $x^2 - x + 2 = 0 \Rightarrow x = -1, 2$

Solⁿ: $a_n = \alpha_1 2^n + \alpha_2 (-1)^n$

To find α_1 and α_2 use a_0 and a_1

Ex-2 $a_n = 6a_{n-1} - 9a_{n-2}$ with $a_0 = 1, a_1 = 6$

Solⁿ $x^2 - 6x + 9 = 0 \Rightarrow x = 3, 3$

$a_n = \alpha_1 3^n + \alpha_2 (n 3^n)$

If equation have same roots then

$$a_n = \alpha_1 3^n + \alpha_2 n 3^n + \alpha_3 n^2 3^n + \alpha_4 n^3 3^n$$

→ The generating function for the sequence $a_0, a_1, \dots, a_k$ of real nos in infinite series,

$$G(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_k x^k = \sum_{k=0}^{\infty} a_k x^k$$

Ex → The generating function for $\{a_k\}$ with $a_k = 3$ and $a_k = 2^k$ are $\sum_{k=0}^{\infty} 3x^k$ and $\sum_{k=0}^{\infty} 2^k x^k$

Generating function for the sequence 1, 1, 1, 1, 1, ...

Ans $\rightarrow 1 + x + x^2 + x^3 + x^4 + \dots$

$\rightarrow$ If $a_k = C(m, k)$ then $G(x) = C(m, 0) + C(m, 1)x + C(m, 2)x^2 + \dots + C(m, m)x^m$

$\rightarrow f(x) = \frac{1}{(1-ax)}$ is generating fn for $1 + ax + a^2x^2 + \dots$

$\rightarrow$ Let 'u' be a real no and 'k' a non-negative integer, then the extended binomial coefficient $\binom{u}{k}$ is defined as

$$\binom{u}{k} = \begin{cases} \frac{u(u-1)(u-2)\dots(u-k+1)}{k!} & \text{if } k > 0 \\ 1 & \text{if } k = 0 \end{cases}$$

Ex ① $\binom{-2}{3} \Rightarrow u = -2 \quad k = 3 \quad \text{so, } \frac{(-2)(-3)(-4)}{3!} = -4$

$$\binom{1/2}{3} = \frac{(1/2)(1/2-1)(1/2-2)}{3!} = 1/16$$

Useful Generating functions

$$(1+x)^n = 1 + C(n, 1)x + C(n, 2)x^2 + \dots + x^n = \sum_{k=0}^n \frac{C(n, k)x^k}{a_k}$$

$$(1+ax)^n = 1 + C(n, 1)ax + C(n, 2)a^2x^2 + \dots + a^n x^n = \sum_{k=0}^n \frac{C(n, k)a^k x^k}{a_k}$$

$$(1+x^r)^n = 1 + C(n, 1)x^r + C(n, 2)x^{2r} + \dots + x^{rn} = \sum_{k=0}^n C(n, k)x^{rk}$$

$$\frac{1-x^{n+1}}{1-x} = 1 + x + x^2 + \dots + x^n = \sum_{k=0}^n x^k$$

$$\frac{1}{1-x} = 1 + x + x^2 + \dots = \sum_{k=0}^{\infty} x^k$$

$$\frac{1}{1-ax} = 1 + ax + a^2x^2 + \dots = \sum_{k=0}^{\infty} a^k x^k$$

$$\frac{1}{1-x^r} = 1 + x^r + x^{2r} + \dots = \sum_{k=0}^{\infty} x^{rk}$$

$$\frac{1}{(1-x)^2} = 1 + 2x + 3x^2 + \dots = \sum_{k=0}^{\infty} (k+1)x^k$$

$$\frac{1}{(1+x)^n} = 1 - C(n,1)x + C(n+1,2)x^2 - \dots = \sum_{k=0}^{\infty} C(n+k-1, k) (-1)^k x^k$$

↳ Alternate (+ and -)

$$\frac{1}{(1-x)^n} = 1 + C(n,1)x + C(n+1,2)x^2 + \dots = \sum_{k=0}^{\infty} C(n+k-1, k) x^k$$

↳ All positive

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

$$\ln(1+x) = \frac{x}{1} - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} x^k$$

$$\rightarrow G(x) \text{ for } d \neq 3 = \frac{1}{1-x}$$

$$G(x) \text{ for } n = \frac{x}{(1-x)^2}$$

$$\text{If } f(x) = \sum_{k=0}^{\infty} a_k x^k \text{ and } g(x) = \sum_{k=0}^{\infty} b_k x^k \text{ then,}$$

$$f(x) + g(x) = \sum_{k=0}^{\infty} (a_k + b_k) x^k, \quad f(x)g(x) = \sum_{k=0}^{\infty} \left(\sum_{j=0}^k a_j b_{k-j} \right) x^k$$

$$\Rightarrow nC_r = \frac{n!}{r!(n-r)!}, \quad \boxed{-nC_r = (-1)^r \frac{(n+r-1)!}{r!(n-1)!} = n+r-1 C_r}$$

$$(1+x)^u = \sum_{k=0}^{\infty} \binom{u}{k} x^k$$

$$\text{So, } (1+x)^{-n} = \sum_{k=0}^{\infty} \binom{-n}{k} x^k = \sum_{k=0}^{\infty} (-1)^k C(n+k-1, k) x^k$$

$$(1-x)^{-n} = \sum_{k=0}^{\infty} C(n+k-1, k) x^k$$

To move from co-ordinate (x_1, y_1) to (x_2, y_2)

$$\rightarrow \text{No. of derangement } D_n = \sum_{i=0}^n (-1)^i \frac{n!}{i!}$$